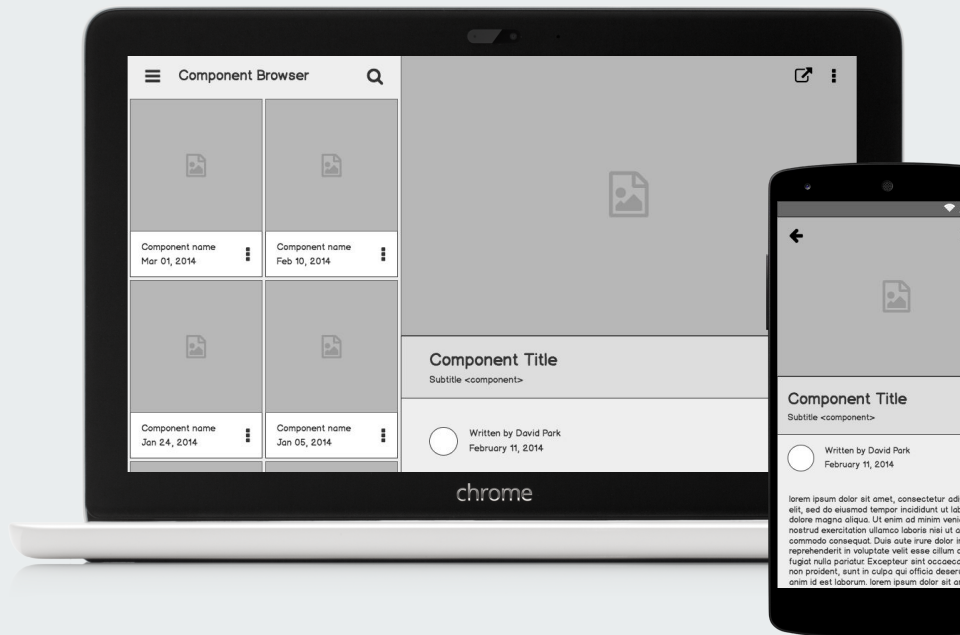




Architecture Challenge

Architecture and Implementation Review





Instructions to run

Prerequisites: Docker Engine with Compose support.

1. Copy ``.env.example`` to ``.env``.

2. Set ``SERIES_HUGGINGFACE_API_KEY`` in ``.env``. This key is required for communication with the Hugging Face provider. Without it, the application starts and the other features remain available, but AI insight requests use only the deterministic fallback and do not call the external AI provider.

3. Start the complete stack:

bash

```
docker compose build --no-cache && docker compose up -d
```

cmd

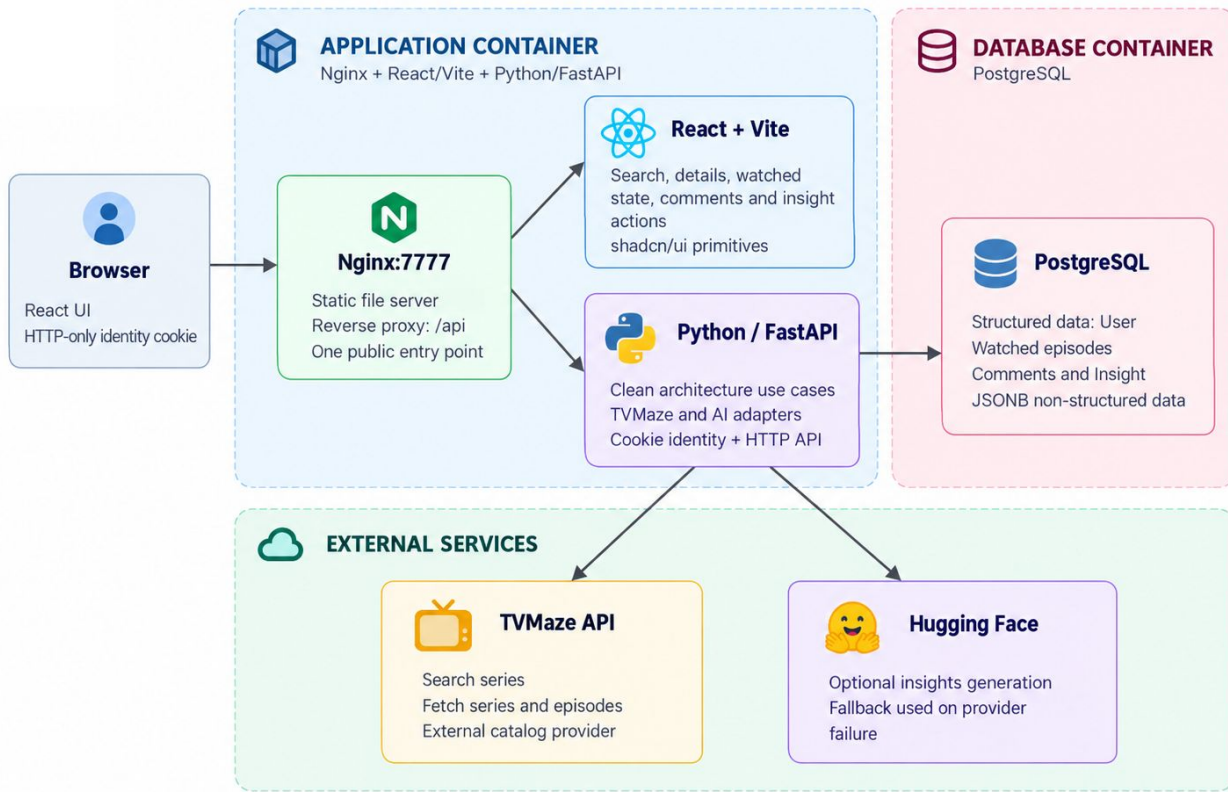
```
docker compose build --no-cache && docker compose up -d
```

powershell

```
docker compose build --no-cache; if ($LASTEXITCODE -eq 0) {  
    docker compose up -d }
```

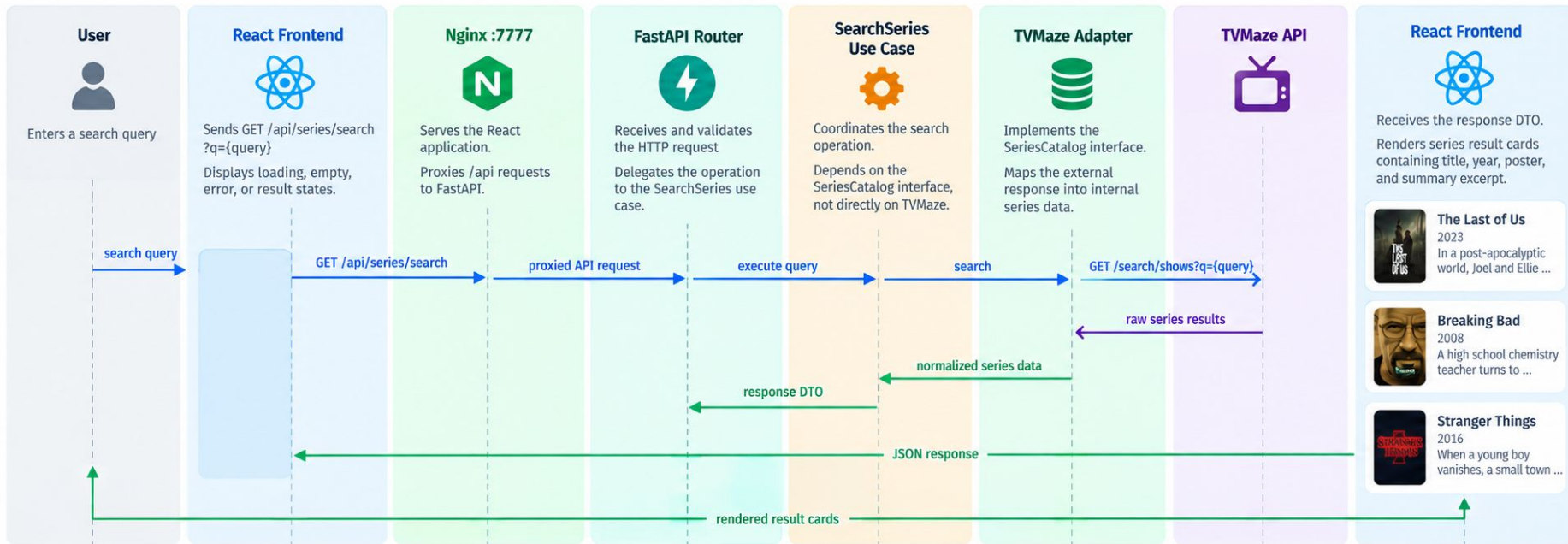
System Architecture

Containers, frontend, backend, database, and external services



TV Series Search — Information Flow

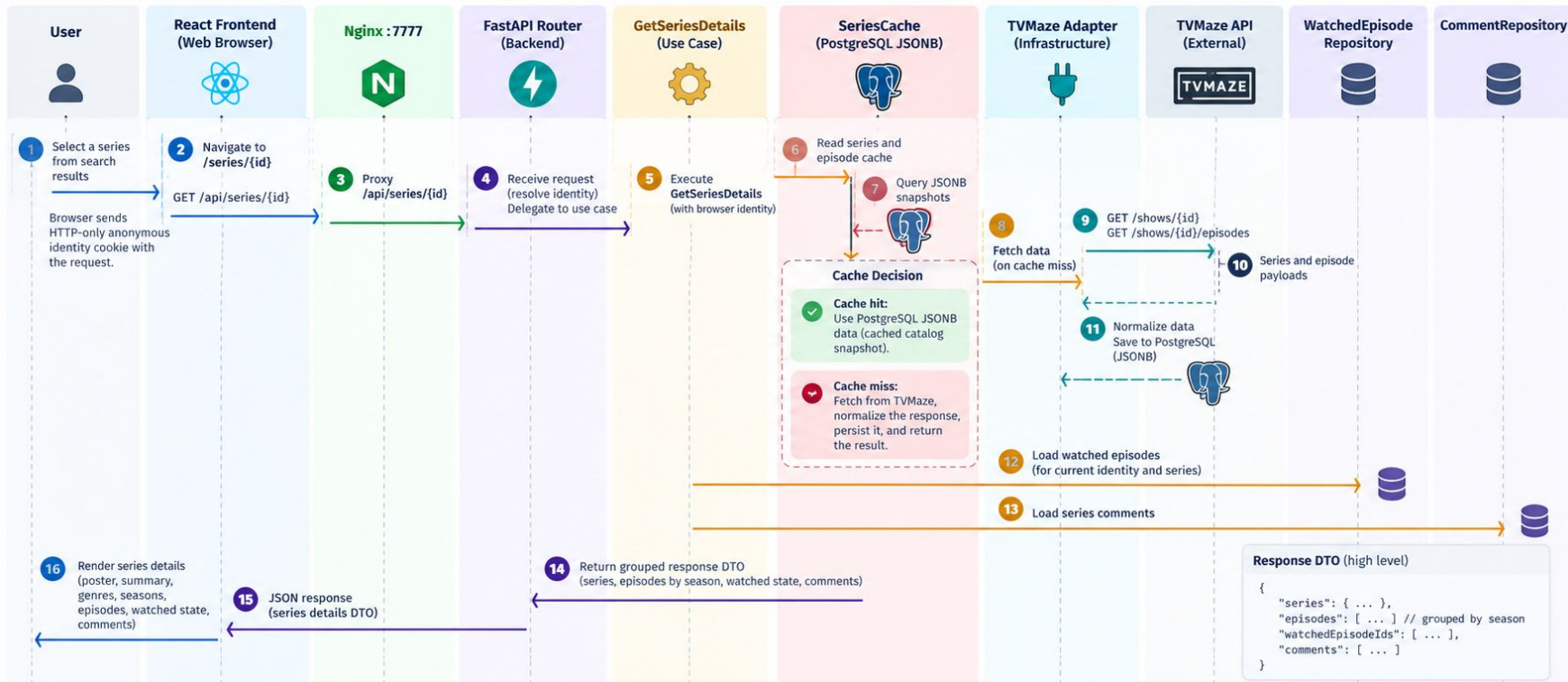
From entering a search query to rendering series results in the frontend.



Series Details — Information Flow

From selecting a series to rendering the full details with episodes, watched state and comments.

The browser sends an HTTP-only anonymous identity cookie with the request to identify the user (per browser).



AI Insight Solution

Optional AI-generated insights for series and episodes, with safe fallback behavior



Input

Information used to generate insights



Series or episode
summary



Genres

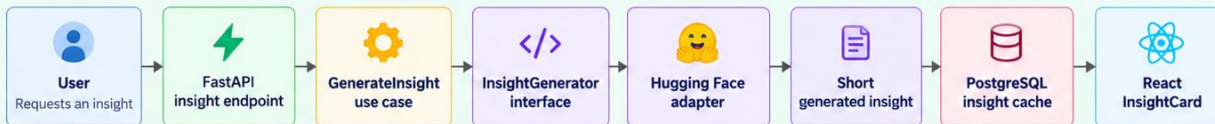


Optional user
comments



Main flow

From user request to generated insight and presentation



Key architectural decisions

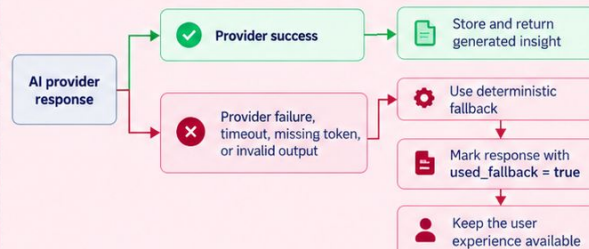
Design principles for a resilient and maintainable solution

- 1 AI integration is isolated behind the `InsightGenerator` interface.
- 2 The application layer does not depend directly on Hugging Face.
- 3 The Hugging Face token remains server-side.
- 4 A fallback provider prevents external AI availability from becoming a single point of failure.
- 5 The same flow supports both series-level and episode-level insights.



Failure strategy

Safe fallback behavior when the AI provider is unavailable



“AI is treated as an optional external capability, not as a dependency of the core business logic.”